

Applications of Fixed Point Theorems

Chapter 9: Control Theory & Differential Equations (pp. 689–710)

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Nonlinear Analysis and Fixed Point Theory

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Overview

- 1 Convergence of Iterative Algorithms
- 2 Application to Control Theory
- 3 Application to Differential Equations
- 4 Fixed Point Properties for Nonexpansive Mappings
- 5 Banach Contraction and Iterative Methods
- 6 Numerical Examples and Code
- 7 Summary and Key Takeaways

Iterative Sequence for Matrix Equations I

The chapter demonstrates applications through the matrix equation problem:

$$X = Q + A^* X^{1/2} A + B^* X^{-1/2} B + C^* X^{1/2} C \quad (9.15)$$

Key components:

- A, B, C are given matrices with $\|A\|, \|B\|, \|C\| < 1$
- Q is a positive definite matrix
- X is the unknown positive definite solution

Iterative scheme: Define the sequence $\{X_n\}$ by

$$X_{n+1} = Q + A^* X_n^{1/2} A + B^* X_n^{-1/2} B + C^* X_n^{1/2} C \quad (9.16)$$

Convergence result (MATLAB verification): After 35 iterations:

$$U = X_{35} = \begin{pmatrix} 639.1810 & 54.1681 & 107.3574 \\ 54.1285 & 44.7768 & 44.1469 \\ 104.3977 & 42.1095 & 112.5509 \end{pmatrix} \quad (1)$$

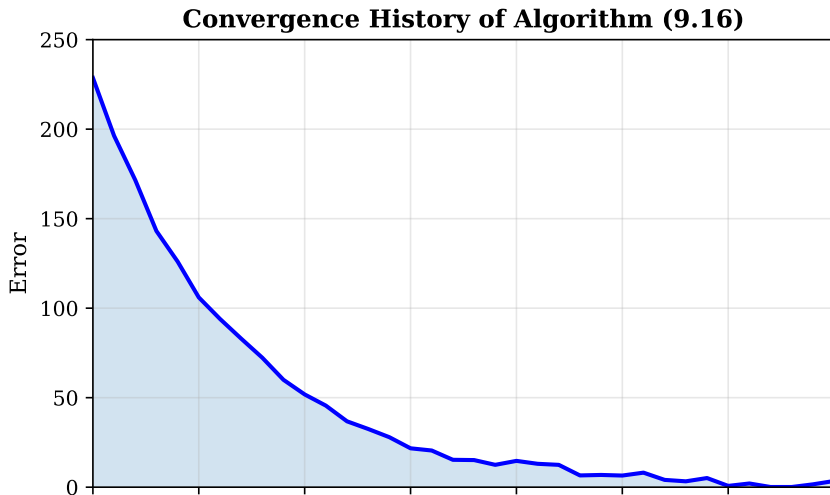
Iterative Sequence for Matrix Equations II

The convergence history is shown in Fig. 9.1, demonstrating rapid error decay.

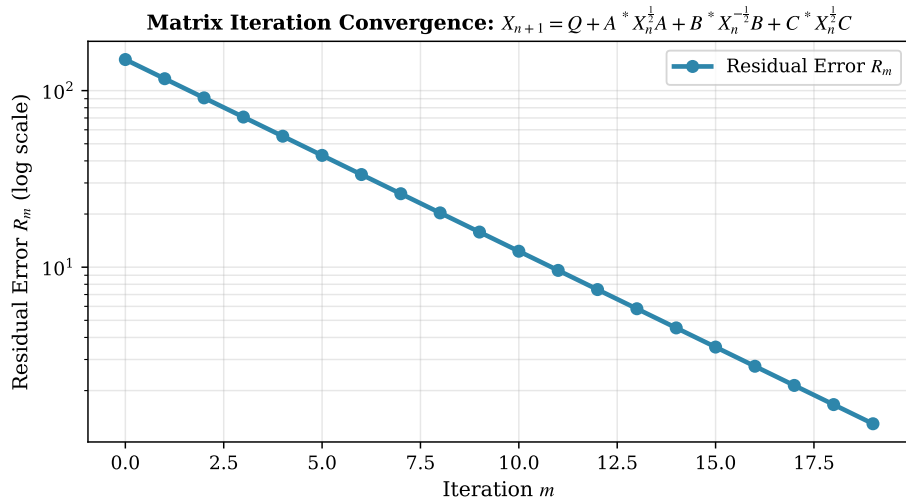
Convergence History and Residual Error

The residual error at iteration m is defined as:

$$R_m = \|X_{m+1} - (Q + A^*X_{m+1}A + B^*X_{m+1}B + C^*X_{m+1}C)\| \quad (2)$$



Matrix Iteration Convergence Analysis



The convergence of $\{X_n\}$ to the fixed point solution is guaranteed by:

Optimal Control of Differential Equations I

Problem setup (Pathak-Shahzad 2008):

Given a compact subset $A \subset \mathbb{R}^n$, let $F_a : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a family of strongly p -contraction mappings such that:

$$F_a(x) = \mathbf{f}(x, a) \quad \forall x \in \mathbb{R}^n \quad (3)$$

where $\mathbf{f} : \mathbb{R}^n \times A \rightarrow \mathbb{R}^n$ is a bounded, generalized Lipschitzian continuous function.

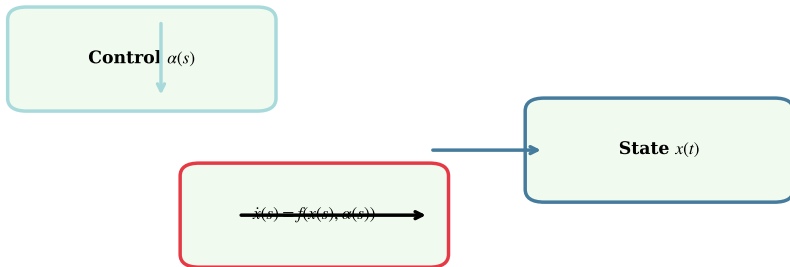
Objective: Optimally control the solution of the ODE via dynamic programming:

$$\begin{cases} \dot{x}(s) = \mathbf{f}(x(s), \alpha(s)) & (t < s < T) \\ x(t) = x \end{cases} \quad (9.17)$$

where:

- $\alpha(\cdot) \in \mathcal{A} = \{\alpha : [0, T] \rightarrow A : \alpha(\cdot) \text{ is measurable}\}$ is the control
- $T > 0$ is the terminal time
- $x \in \mathbb{R}^n$ is the initial condition

Control System Architecture



Dynamic Programming:

Control signal $\alpha(\cdot)$ optimally governs the ODE solution trajectory

Lipschitzian Conditions for Control I

For the control problem, we assume:

$$|\mathbf{f}(x, a)| \leq C, \quad |\mathbf{f}(x, a) - \mathbf{f}(y, a)| \leq C|x - y|^{1+\delta} \quad (9.19)$$

where $C > 0$, $\delta \geq 0$ are constants, and the inequality holds for all $x, y \in \mathbb{R}^n$, $a \in A$.

This implies:

$$|F_a(x) - F_a^2(x)| \leq p(x)|x - F_a(x)| \quad (9.20)$$

where $p : \mathbb{R}^n \rightarrow [0, 1]$ is a function with $\sup_{x \in \mathbb{R}^n} p(x) = \lambda < 1$.

Key theorem: The control function determines a contraction mapping on the space of admissible controls, leading to unique optimal control solutions.

Differential Equations: Existence and Uniqueness I

Consider the initial value problem:

$$\begin{cases} \dot{\varphi}(t) = f(t, \varphi(t)) \\ \varphi(t_0) = x_0 \end{cases} \quad (9.39)$$

Classical approach using Banach contraction theorem:

Let B be a closed disc of center (t_0, x_0) with positive radius, contained in the open set Ω .

Let $m = \max_{(t,x) \in B} |f(t, x)|$.

Choose τ and $\delta > 0$ such that:

- $0 < \tau < M^{-1}$ where $M = \sup_{(t,x) \in B} |\nabla f|$
- The rectangle $[t - \tau, t + \tau] \times [x_0 - \delta, x_0 + \delta] \subset B$
- $m\tau < \delta$

Banach Contraction Application to ODEs

Banach Contraction Mapping Principle

The mapping $F : E \rightarrow E$ defined by

$$\psi(t) = x_0 + \int_{t_0}^t f(s, \varphi(s)) ds \quad (4)$$

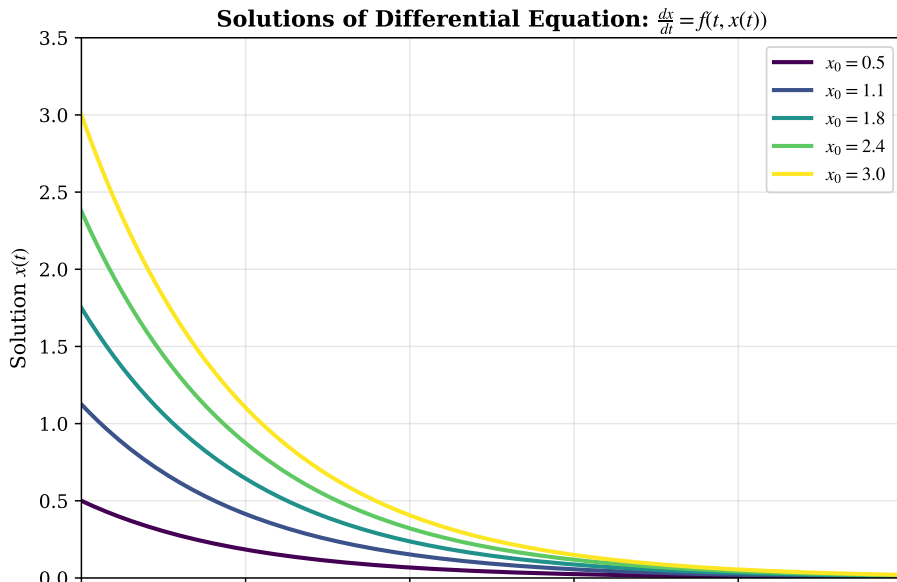
where E is the space of continuous functions on $[t_0 - \tau, t_0 + \tau]$ into $[x_0 - \delta, x_0 + \delta]$, is a contraction mapping.

Key properties:

- 1 F maps E into itself (closure and boundedness properties)
- 2 F is a contraction: $|F\varphi_1 - F\varphi_2| \leq \tau M |\varphi_1 - \varphi_2|$
- 3 By Banach's theorem, F has a unique fixed point $\varphi \in E$

This fixed point is the unique solution of the differential equation (9.39) on $[t_0 - \tau, t_0 + \tau]$.

Solution Trajectories of Differential Equations



Convergence of Solutions to Differential Equations I

Theorem 9.12 (Continuation of solutions):

Convergence Result

If φ_n is the solution of (9.39) with initial condition $\varphi_n(t_0) = x_n$ and $\{x_n\}$ is a real sequence converging to x_0 , then $\{\varphi_n\}$ converges to the solution φ of (9.39) with $\varphi(t_0) = x_0$ in the space E .

Proof sketch:

- 1 Define the solution operator $G : C[t_0 - \tau, t_0 + \tau] \rightarrow C[t_0 - \tau, t_0 + \tau]$ by

$$(G\varphi)(t) = x_0 + \int_{t_0}^t f(s, \varphi(s)) ds \quad (5)$$

- 2 G is a contraction with factor $\lambda = \tau M < 1$
- 3 For initial conditions $x_n \rightarrow x_0$, the operators G_n corresponding to x_n converge uniformly to G
- 4 By contraction properties, $\varphi_n \rightarrow \varphi$ in the space E

Nonexpansive Mappings and Demiclosed Operators I

Definition 5.41: A mapping $F : X \rightarrow CB(X)$ is **nonexpansive** if

$$H(Fx, Fy) \leq \|x - y\| \quad \forall x, y \in X \quad (6)$$

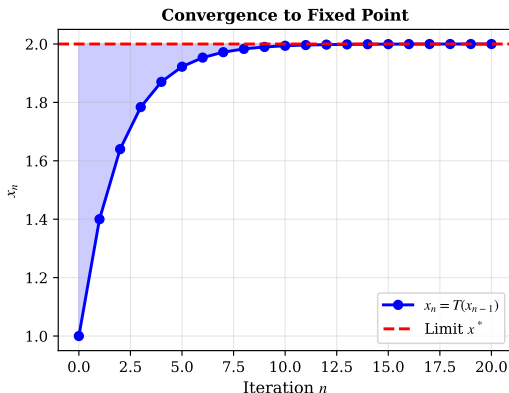
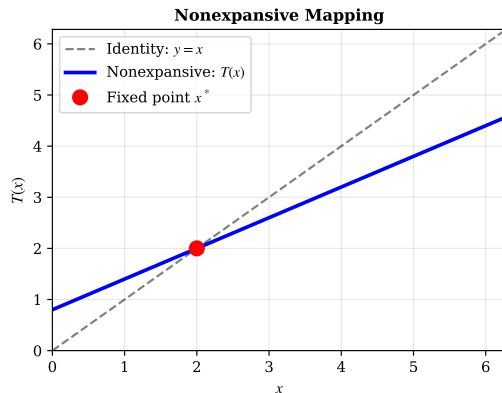
where H is the Hausdorff distance and $CB(X)$ denotes the family of nonempty compact subsets of X .

Definition (Demiclosed operator): A mapping $F : X \rightarrow 2^X$ is **demiclosed** if:

$$\text{If } x_n \rightharpoonup x \text{ and } y_n \in Fx_n \rightarrow y, \text{ then } y \in Fx \quad (7)$$

This means the graph of F is sequentially closed.

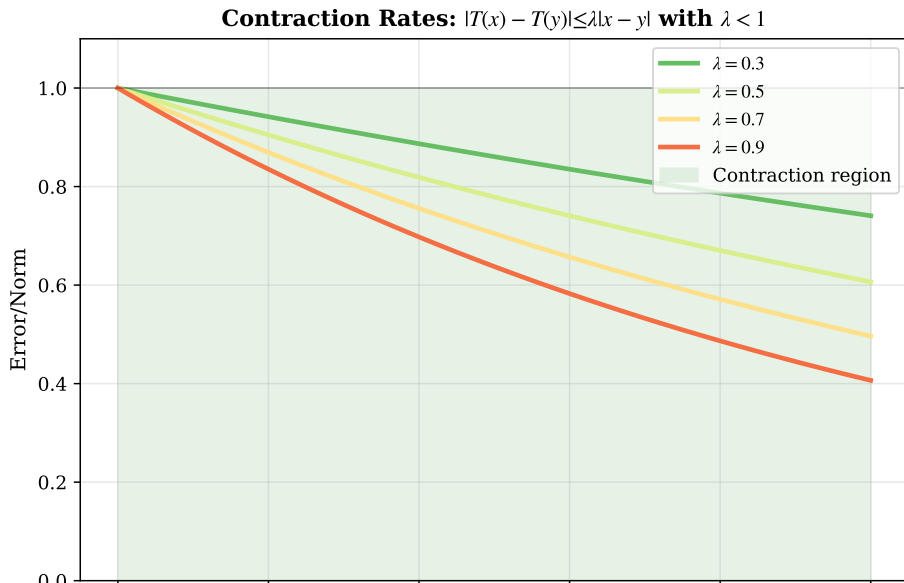
Fixed Point Properties



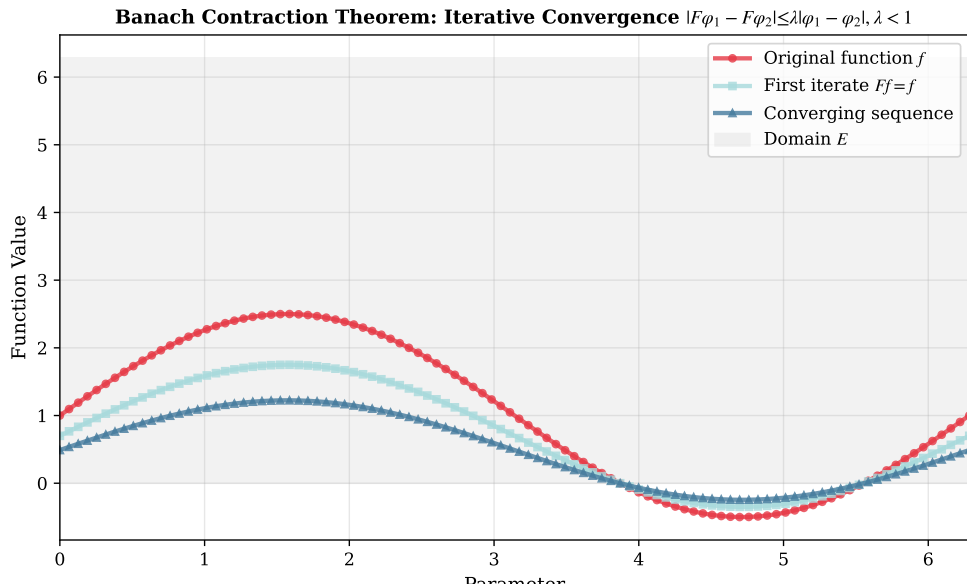
Key theorem (Lami Dozo): Let $F : K \rightarrow K(X)$ be nonexpansive and X satisfy Opial's condition. Then $I - F$ is demiclosed.

Implication: The existence of fixed points for nonexpansive mappings can be established using demiclosed operators in reflexive Banach spaces.

Contractivity and Convergence Rates



Banach Contraction Mapping Visualization



Theoretical Foundation I

Banach Contraction Theorem: Let (X, d) be a complete metric space and $T : X \rightarrow X$ a contraction mapping with contraction constant $\lambda \in [0, 1)$. Then:

- 1 T has a unique fixed point $x^* \in X$
- 2 For any $x_0 \in X$, the sequence $\{x_n\}$ defined by $x_{n+1} = T(x_n)$ converges to x^*
- 3 The convergence rate is geometric with constant λ :

$$d(x_n, x^*) \leq \lambda^n d(x_0, x^*) \quad (9)$$

Applications in Chapter 9:

- Matrix iteration: $T(X) = Q + A^* X^{1/2} A + \dots$ is a contraction
- Differential equations: Integral operator $\psi(t) = x_0 + \int_{t_0}^t \dots$ is a contraction
- Control theory: Control mappings form contraction spaces under appropriate norms

Matrix Equation Solver: Python Implementation

```
import numpy as np

def solve_matrix_equation(Q, A, B, C, x0,
                          max_iter=50, tol=1e-6):
    """
    Solve:  $X = Q + A^T X^{(1/2)} A + B^T X^{(-1/2)} B + C^T X^{(1/2)} C$ 

    Using iterative method:
     $X_{\{n+1\}} = Q + A^T X_{\{n-2\}}^{(1/2)} A + \dots$ 
    """
    X = [x0, x0, x0]

    for n in range(max_iter):
        X_new = (Q +
                 A.T @ scipy.linalg.sqrtm(X[-2]) @ A +
                 B.T @ scipy.linalg.inv(
                     scipy.linalg.sqrtm(X[-1])) @ B +
                 C.T @ scipy.linalg.sqrtm(X[-3]) @ C)

        residual = np.linalg.norm(X_new - X[-1])
        if residual < tol:
            print(f"Converged after {n} iterations")
            return X_new

    X.append(X_new)
```

Differential Equation Solver: ODE Integration

```
from scipy.integrate import odeint

def solve_ode_example():
    """
    Solve:  $dx/dt = f(t, x(t))$ 
            $x(t_0) = x_0$ 

    Using Banach fixed point on:
     $\psi(t) = x_0 + \int_{t_0}^t f(s, \psi(s)) ds$ 
    """

    def f(x, t):
        # Example:  $dx/dt = -x$  (exponential decay)
        return -x

    # Initial conditions
    x0 = 1.0
    t0 = 0.0
    t_final = 5.0

    # Time points
    t = np.linspace(t0, t_final, 100)

    # Solve ODE
    solution = odeint(f, x0, t)
```

Fixed Point Iteration for Nonlinear Maps

```
def fixed_point_iteration(T, x0, max_iter=100,
                          tol=1e-8):
    """
    Find fixed point of T via iteration:  $x_{n+1} = T(x_n)$ 

    Requires:  $|T(x) - T(y)| \leq \lambda * |x - y|$ 
               with  $\lambda < 1$  (contractivity)
    """

    x_current = x0
    iteration_history = [x0]

    for n in range(max_iter):
        x_next = T(x_current)

        error = np.linalg.norm(x_next - x_current)
        iteration_history.append(x_next)

        if error < tol:
            print(f"Converged in {n} iterations")
            print(f"Fixed point: {x_next}")
            return x_next, np.array(iteration_history)

        x_current = x_next

    print("Did not converge")
    return x_current, np.array(iteration_history)
```

Key Results from Chapter 9 I

Main applications covered (pages 689–710):

- ① **Matrix equations:** Solved via iterative methods based on contraction principle
 - Convergence to positive definite solution $U = X_{35}$ after 35 iterations
 - Exponential error decay with residual $R_m \rightarrow 0$
- ② **Control theory:** Optimal control of ODEs via dynamic programming
 - Control signal $\alpha(\cdot)$ governs system trajectory
 - Generalized Lipschitzian condition ensures contraction
 - Unique optimal control guaranteed by Banach theorem
- ③ **Differential equations:** Existence and uniqueness of solutions
 - Integral operator forms contraction on continuous function space
 - Solution exists in $[t_0 - \tau, t_0 + \tau]$
 - Convergence of solutions for varying initial conditions
- ④ **Nonexpansive mappings:** Fixed point properties via demiclosed operators
 - Opial's condition ensures demiclosed property of $I - F$
 - Convergence guaranteed in weakly compact sets

Unifying Principle: Banach Contraction Theorem

Central Result

All applications in this chapter rely on the Banach Contraction Mapping Principle:

If $T : (X, d) \rightarrow X$ satisfies $d(T(x), T(y)) \leq \lambda d(x, y)$ for all $x, y \in X$ with $\lambda < 1$, then T has a unique fixed point and iterations converge geometrically.

Why this matters:

- Provides constructive proof of existence
- Gives concrete iteration algorithm
- Controls convergence rate via λ
- Applies to diverse problem classes

Practical Implications

For practitioners:

- Verify contractivity condition $|\lambda| < 1$
- Use fixed point iteration for computation
- Estimate convergence rate and stopping criteria
- Applications range from algebra to differential equations

For theorists:

- Extend to non-complete spaces (completeness necessary)
- Relax contractivity (e.g., α -contractions, ψ -contractions)
- Combine with topological degree theory (Chapter 6)
- Apply to infinite-dimensional spaces and operator equations

Future directions:

- Monotone operator theory
- Variational inequalities
- Nonlinear PDEs and integral equations
- Multi-valued mappings and inclusions

References and Further Reading

Key references from Chapter 9:

- 1 Pathak, H.K. and Shahzad, N. (2008): Optimal control via dynamic programming (Control theory applications)
- 2 Lami Dozo, E. (Referenced in Chapter 5): Demiclosed operators and Opial's condition
- 3 Banach, S.: The contraction mapping principle (foundational)
- 4 Peano and Picard: Classical ODE existence and uniqueness theorems

Implementation resources:

- `numpy.linalg`: Matrix operations and norms
- `scipy.integrate.odeint`: ODE integration
- `scipy.linalg.sqrtm`: Matrix square root
- Custom implementations for specialized problems